

FINAL PROJECT REPORT

Medical Diagnosis Center by ServiceNow Service Portal

Platform: ServiceNow | Date: 06 October 2026

1. INTRODUCTION

The Medical Diagnosis Center by ServiceNow Service Portal is a healthcare-service workflow prototype designed around a patient-facing diagnostic booking and reporting journey. The implemented experience begins with the Medical Test Catalog and continues through diagnostic-test selection, Book Now, Patient Details, Schedule, Payment, Done, Appointment, My Appointments, Appointment Completed, automatic Report creation, Report and My Lab Reports.

1.2 Purpose

Provide a structured diagnostic-test booking experience, reduce manual coordination, standardize patient/test/schedule data, provide appointment and report visibility, and protect patient information through configured roles and ACLs.

2. IDEATION PHASE

2.1 Problem Statement

Patients need a faster, clearer and more traceable way to discover diagnostic tests, book appointments and access completed reports. A fragmented process can create incomplete information, scheduling confusion, repeated coordination and poor visibility into appointment/report status.

2.2 Empathy Map Canvas

Stakeholder	Think & Feel	See	Say & Do	Pain	Gain
Patient / Service Requester	Wants simple booking and clear status.	Medical Test Catalog, booking screens, My Appointments, My Lab Reports.	Selects test, books, tracks appointment and views reports.	Repeated coordination and unclear status.	Structured booking and visibility.
Service Administrator	Wants maintainable automation.	Service Portal, Flow Designer, records, ACLs.	Configures workflow and security.	Configuration/debugging effort.	Centralized control and execution visibility.
Authorized Healthcare Staff	Wants sufficient appointment context.	Appointment and status information.	Performs authorized service actions.	Incomplete context.	Cleaner handoff.
Diagnostic / Laboratory Team	Wants complete test and appointment information.	Appointment and report records.	Performs diagnostic-service fulfillment.	Incomplete data.	Standardized operational data.

2.3 Brainstorming

Guided Medical Test Catalog; diagnostic-test selection; Book Now; Patient Details; Schedule; configured Payment → Done; Appointment tracking; My Appointments; Appointment Completed; automatic Report creation; My Lab Reports; role/ACL protection. The selected concept is the ServiceNow diagnostic booking-to-report lifecycle.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	Requester Experience	System Response
Discover Test	Patient opens the Medical Test Catalog and identifies the required diagnostic test.	ServiceNow displays configured diagnostic test options.
Select Test	Patient selects a diagnostic test.	Selected test context is carried into the booking journey.
Book Now	Patient chooses Book Now.	The booking flow opens the configured patient-detail stage.
Patient Details	Patient enters required patient/request information.	Configured form fields and validation capture the required details.
Schedule	Patient selects the available/configured appointment date and time.	The schedule selection is retained for the appointment.
Payment	Patient reaches the configured payment step.	The configured payment stage is processed within the prototype scope.
Done	Patient completes the booking journey.	The booking is finalized and the Appointment becomes available.
Appointment	Patient/authorized staff access the created appointment.	Appointment details and lifecycle status are available.
My Appointments	Patient views booked appointments.	My Appointments provides the configured appointment list/status.
Appointment Completed	Authorized staff complete the appointment.	The completion event updates the configured lifecycle.
Auto Create Report	Completed appointment triggers report creation.	Configured automation creates the Report record.
Report	Authorized user accesses the generated report.	The configured Report information is available.
My Lab Reports	Patient views available lab reports.	My Lab Reports exposes the configured report for the patient.

3.2 Solution Requirements

Requirement	Description
FR-1	Provide a Medical Test Catalog through the ServiceNow Service Portal.
FR-2	Allow the requester to select a configured diagnostic test and start the Book Now journey.
FR-3	Capture required patient/requester details using configured portal fields and validation.
FR-4	Support appointment date/time selection through the Schedule stage.
FR-5	Support the configured Payment → Done booking stage without claiming an external payment gateway unless implemented.
FR-6	Create the Appointment with the selected test, patient and schedule context.
FR-7	Provide My Appointments for viewing the configured appointment lifecycle.
FR-8	Support Appointment Completed as the configured completion event.
FR-9	Automatically create the Report after the configured appointment-completion event.
FR-10	Provide Report and My Lab Reports visibility to authorized users.
FR-11	Protect patient/appointment/report records with configured roles and ACLs.

Requirement	Description
FR-12	Support troubleshooting and validation through Flow Designer Execution Details and platform logs.

3.3 Data Flow Diagram

Medical Test Catalog → Select Diagnostic Test → Book Now → Patient Details → Schedule → Payment → Done → Appointment → My Appointments → Appointment Completed → Auto Create Report → Report → My Lab Reports.

3.4 Technology Stack

Component	Technology	Purpose
User Interface	ServiceNow Service Portal / Medical Test Catalog	Patient-facing diagnostic-test discovery and booking experience.
Catalog / Form	Catalog items, variables, UI Policies / form validation	Patient details and diagnostic-test selection data capture.
Workflow Automation	ServiceNow Flow Designer / configured automation	Booking, appointment lifecycle and automatic report creation.
Appointment	Configured ServiceNow appointment record/process	Stores scheduled diagnostic appointment information.
Report	Configured ServiceNow Report record/process	Stores the report generated after appointment completion.
Patient Views	My Appointments / My Lab Reports	Provides patient-facing appointment and report visibility.
Security	Roles + ACLs	Controlled access to patient, appointment and report information.
Diagnostics	Flow Execution Details / platform logs	Workflow troubleshooting and validation.
Infrastructure	ServiceNow cloud / PDI	Execution and storage environment.

4. PROJECT DESIGN

4.1 Problem Solution Fit

The solution fits the problem by replacing fragmented diagnostic booking and report access with a centralized ServiceNow portal journey. Patients receive guided selection and booking; authorized staff receive appointment/completion context; administrators gain configuration and execution visibility.

4.2 Proposed Solution

The proposed solution combines Medical Test Catalog, Book Now, Patient Details, Schedule, Payment → Done, Appointment, My Appointments, Appointment Completed, automatic Report creation, Report and My Lab Reports within ServiceNow.

4.3 Solution Architecture

Patient → Service Portal / Medical Test Catalog → Booking stages → Appointment → My Appointments → Appointment Completed → Report creation → My Lab Reports, with Flow Designer/configured automation, roles and ACLs supporting the lifecycle.

5. PROJECT PLANNING & SCHEDULING

Sprint	Epic / Focus	Story Points	Release Outcome
Sprint 1	Medical Test Catalog & Booking UI	9	Diagnostic catalog, test selection, Book Now and patient-detail screens.
Sprint 2	Schedule, Payment & Appointment	15	Schedule stage, configured Payment → Done path and Appointment creation.
Sprint 3	My Appointments, Completion & Report	13	Appointment lifecycle, completion event, automatic Report creation and My Lab Reports.
Sprint 4	Testing, Security & Documentation	16	Functional/UAT validation, ACL checks, workflow diagnostics and final documentation.

6. FUNCTIONAL AND PERFORMANCE TESTING

The functional test set validates catalog selection, booking, schedule/payment, appointment lifecycle, automatic report creation, report visibility, security and the complete journey. Performance testing is limited to project-level portal/workflow observations; no production-scale benchmark was supplied.

Test ID	Test Scenario	Final Result
FT-01	Medical Test Catalog and diagnostic-test selection	Pass
FT-02	Book Now and Patient Details validation	Pass
FT-03	Schedule stage and appointment data capture	Pass
FT-04	Payment → Done booking sequence	Pass
FT-05	Appointment creation and data mapping	Pass
FT-06	My Appointments visibility and status	Pass
FT-07	Appointment Completed transition	Pass
FT-08	Automatic Report creation after completion	Pass
FT-09	Report and My Lab Reports visibility	Pass
FT-10	Role and ACL access	Pass
FT-11	Configured notification/status behaviour	Pass
FT-12	End-to-end diagnostic booking-to-report workflow	Pass
PT-01	Portal response behaviour	Pass - no abnormal delay reported in project testing
PT-02	Repeated booking execution consistency	Pass
PT-03	Workflow execution consistency	Pass
PT-04	Portal/form compatibility	Pass

7. RESULTS

Output Area	Implemented Result
Medical Test Catalog	Configured diagnostic test catalogue and selection experience.
Booking Journey	Book Now → Patient Details → Schedule → Payment → Done.

Output Area	Implemented Result
Appointment	Appointment created with configured patient, test and schedule context.
My Appointments	Patient-facing appointment list/status visibility.
Completion Automation	Appointment Completed acts as the configured completion event.
Report	Report record is automatically created after configured completion.
My Lab Reports	Patient-facing access to available lab reports.
Security	Configured roles and ACLs protect operational information.
Traceability	Workflow execution details and record relationships support validation.

8. ADVANTAGES & DISADVANTAGES

Advantages	Disadvantages / Current Limitations
Provides a clear diagnostic-test booking journey.	Current implementation is a project prototype rather than a production clinical information system.
Centralizes test selection, patient details, scheduling and appointment tracking.	Direct hospital/EMR/laboratory/API integration is not implemented.
Automates the transition from appointment completion to report creation.	Advanced SLA/escalation and analytics are future scope.
Provides My Appointments and My Lab Reports for clearer patient visibility.	External SMS/WhatsApp and other messaging channels are future scope.
Uses ServiceNow low-code automation and controlled access.	No production-scale load benchmark figures were supplied.
Does not autonomously diagnose or prescribe; clinical decisions remain with authorized professionals.	Clinical interpretation and treatment decisions are outside the implemented workflow.

9. CONCLUSION

The implemented prototype provides a structured diagnostic booking-to-report workflow in ServiceNow. It improves patient-facing navigation, appointment visibility and report access while maintaining controlled operational access. It does not autonomously diagnose, prescribe or make clinical decisions.

10. FUTURE SCOPE

Future scope includes authorized hospital/EMR/laboratory integrations, richer fulfilment automation, advanced appointment/approval routing, SLA/escalation, external messaging, analytics and governed AI-assisted service classification or anomaly detection.

11. APPENDIX

Functional Data Mapping

Data Element	Source	Destination	Purpose
Patient / Requested For	Patient Details / configured variable	Appointment → Patient / Requested For	Identifies the patient for the booked diagnostic service.
Contact / Mobile	Patient Details	Appointment → Contact	Stores approved requester contact information.
Diagnostic Test	Medical Test Catalog selection	Appointment → Diagnostic Test	Identifies the selected diagnostic test.
Schedule	Schedule stage	Appointment → Appointment Details	Stores the selected date/time information.
Payment Status	Payment stage	Booking / Appointment → Payment Status	Tracks the configured payment-stage outcome.
Appointment Status	Appointment lifecycle	Appointment → Status	Tracks booked, completed or other configured states.
Report	Appointment Completed event	Report record	Stores the report created after completion.
My Lab Reports	Report record	Patient-facing report list	Provides access to available reports.